Nader Khalil

Full-Stack Developer

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PROFESSIONAL SUMMARY

Full-stack software engineer with a B.Sc. in Software Engineering, building web apps and AI assistants engineered for high reliability, strict type safety, and test automation. Proven track record across Next.js, TypeScript, PHP 8.4, and PostgreSQL, from architecture and database row-locking to production CI/CD pipelines.

TECHNICAL SKILLS

Frontend: React 19, Next.js, TypeScript, Vite, TanStack Query, Tailwind CSS, accessible UI (axe)

Backend: REST APIs, layered architecture, PHP 8.4, C# MinimalAPI, Zod, Auth.js

Data & Security: PostgreSQL + pgvector, MariaDB, Prisma, row locks, CSP, CSRF, rate limiting

AI & Automation: Tool calling with human review, RAG, embeddings, prompt caching, Vercel AI SDK

Testing & CI/CD: Playwright, Vitest, PHPUnit, race tests, GitHub Actions, Docker

Languages: TypeScript, JavaScript, Python, C, C++, C#, SQL, Bash, Linux

TECHNICAL PROJECTS

AI Booking Platform 2026

Live build: booking.naderkhalil.dev · Case study: naderkhalil.dev/case-study.html

- Built a configurable booking platform: clients book from live calendars, advisors manage availability, and admins run knowledge bases and rules. (Next.js 16, TypeScript, PostgreSQL + pgvector, Prisma, Auth.js).
- RAG assistant answers from admin-managed docs with citations; tool proposals require explicit user confirmation before booking or cancellation. Verified by 120 live AI safety checks (prompt injection, scope).
- Engineered concurrency-safe bookings with row locks race-tested on PostgreSQL; backed by 766 automated tests (357 unit, 320 integration, 89 e2e Playwright) in GitHub Actions CI.

Student-Faculty Services v2, *Al Ain University capstone rewrite* 2026

Live build: student-faculty-services-v2.infinityfree.io · Case study: naderkhalil.dev/case-study-v2.html

- Architected a framework-free PHP 8.4 REST API (46 endpoints) and React 19 frontend with 10-second cancellation undo, reminders, waitlists, and 0 double bookings across 8-student concurrent race tests.
- Enforced strict layered architecture with 500+ PHPUnit, Vitest, and Playwright tests with 100/100 Lighthouse accessibility.

42 Intra Logtime Calculator, *published Chrome extension* 2026

Chrome Web Store: chromewebstore.google.com

- Lightweight Chrome extension (19 KB, 60+ users, 5.0★) providing real-time calendar logtime calculations client-side with zero data collection across both legacy and V3 42 Intra layouts.

PROFESSIONAL EXPERIENCE

AI Trainer & Mentor, *Department of Education and Knowledge (ADEK)* 07/2026 – 08/2026

- Delivered technical instruction on agentic workflows, prompt structuring, and autonomous tool calling for the accredited AgenticVerse program; mentored teams building AI prototypes for ADNOC, FANR, and Khalifa Fund.
- Guided participants through API schema validation, structured output parsing, and debugging in Python and TypeScript.

QA & Software Tester, *42 Abu Dhabi* 01/2026

- Conducted functional and UI testing on an internal student platform using black-box scenarios; authored reproduction reports with logs and steps to help developers isolate defects quickly.
- Tested features across browsers and viewports; re-tested defect fixes across user flows to ensure zero regressions before release.

IT Support & Technical Staff, *Ticketmaster* 12/2025

- Provisioned and deployed entry ticketing hardware and access scanners for a high-volume international event; maintained 100% operational uptime through real-time hardware health checks.
- Served as on-site technical point of contact, delivering rapid incident resolution and verifying network connectivity under strict SLAs.

Software Engineering Intern, *Timex IT Consulting* 01/2024 – 04/2024

- Developed C# MinimalAPI CRUD endpoints with request validation and standardized HTTP response handling.
- Supported software release rollouts by identifying, documenting, and verifying functional defect fixes on client accounting software.

EDUCATION & CERTIFICATIONS

B.Sc. in Software Engineering, *Al Ain University*

Computer Science Diploma, *42 Abu Dhabi*

MOEI × 42 Abu Dhabi AgentEra Hackathon, *Honoree*

CyberQ CTF, Top 10 Finalist, *UAE Cybersecurity Council*

Hack The Box University CTF, *Top 12% (119/1,014 teams)*

RecognAIze Hackathon, *AI on Watch with Datamind*

Languages: Arabic (Native), English (Professional Proficiency)

Portfolio & Case Studies: naderkhalil.dev